PRACTICAL COURSE WORKBOOK

AI in Healthcare & Diagnostics

Work with digital-health information responsibly

LEVEL	Intermediate
GUIDED TIME	5 hours across 12 lessons
PROJECT	a safe, fictional health-data workflow and review
SKILLS	Medical AI, Imaging, Clinical NLP, FDA AI, Predictive models, Critical evaluation
EDITORIAL STATUS	Structured draft v0.9

WHAT YOU WILL BE ABLE TO DO

- Use Medical AI appropriately in a realistic, bounded task.
- Use Imaging appropriately in a realistic, bounded task.
- Use Clinical NLP appropriately in a realistic, bounded task.
- Use FDA AI appropriately in a realistic, bounded task.

WHO THIS IS FOR

Learners exploring health technology without replacing clinical expertise.

BEFORE YOU BEGIN

- Basic data literacy

COURSE MAP

From foundation to finished work

AI in Healthcare & Diagnostics is a structured, practical course covering Medical AI, Imaging, Clinical NLP, FDA AI, Predictive models. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Medical AI: Health context	1. Care pathways with Medical AI 2. Health data with Imaging 3. Stakeholders with Clinical NLP
02 Imaging: Digital systems	1. Standards with FDA AI 2. Interoperability with Predictive models 3. Workflows with Medical AI
03 Clinical NLP: Quality and safety	1. Validation with Imaging 2. Privacy with Clinical NLP 3. Human oversight with FDA AI
04 FDA AI: Applied evaluation	1. Implementation with Predictive models 2. Monitoring with Medical AI 3. Equity with Imaging

HOW TO STUDY EACH LESSON

1. Read the objective and identify the decision it supports.
2. Reproduce the worked example with the stated input.
3. Test one normal case and one boundary or failure case.
4. Complete the knowledge check without guessing.
5. Save a short evidence note: result, limitation and next action.

CAPSTONE PROJECT

Produce a reviewable AI in Healthcare & Diagnostics project using Medical AI, Imaging, Clinical NLP.

DELIVERABLE	A working AI in Healthcare & Diagnostics artefact plus an evidence-based self-review.
TOOLS	A suitable Medical AI environment, A plain-text decision log, Test data or realistic sample material

Production plan

1. Define the intended user, outcome and constraints.
2. Create the smallest complete result using Medical AI.
3. Test one normal case, one boundary case and one failure response.
4. Revise the work from the evidence and preserve before-and-after results.
5. Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

RESPONSIBLE PRACTICE

This material is educational and does not provide medical advice or replace qualified clinical judgement.

Evidence log

INPUT	Original brief, data, requirements or test material.
DECISIONS	Chosen method and one reasonable alternative.
TESTS	Normal, boundary and failure results.
LIMITATION	What the result does not establish.
REVISION	One evidence-led improvement.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

Authoritative references

Global strategy on digital health 2020-2027

World Health Organization - reviewed 2026-08-21
https://www.who.int/publications/i/item/9789240116870

FHIR specification

HL7 International - reviewed 2026-08-21
https://hl7.org/fhir/

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Medical AI scope.